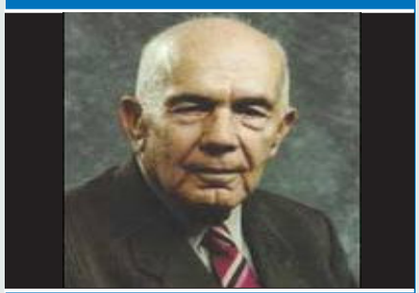
10 Operations Research



Frank Lauren Hitchcock
(1875-1957)

Introduction

Operations reserach (O.R.) is an analytical method of problem solving and decision-making, that is useful in management organisations. The transportation problem involves certain origins (sources) which may represent factories where we produce homogeneous items and a number of destinations where we supply a required quantity of the products. Each factory has a certain capacity constraint and each destination (dealer or customer) has a certain requirement. The unit cost of transportation of the items from the factory to the dealer/ customer is known. American mathematician and physicist Frank Lauren Hitchcock (1875-1957) known for his formulation of transportation problem in 1941.

Learning Objectives

After studying this chapter students are able to understand

- formulate the transportation and assignment problems
- distinguish between transportation and assignment problems
- find an initial basic feasible solution of a transportation problem
- identify the degeneracy and non-degeneracy in a transportation problem
- find the solution of an assignment problem by Hungarian method.
- distinguish between tactic and strategic decisions
- find the best alternatives using maximin and minimax criteria

10.1 Transportation Problem

The objective of transportation problem is to determine the amount to be transported from each origin to each destinations such that the total transportation cost is minimized.

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10.1.1 Definition and formulation

The Structure of the Problem

Let there be m origins and n destinations. Let the amount of supply at the ith origin is a_i. Let the demand at j^{th} destination is b_j.

The cost of transporting one unit of an item from origin i to destination j is c_{ij} and is known for all combinations (i,j). Quantity transported from origin i to destination j be x_{ij}

The objective is to determine the quantity x_{ij} to be transported over all routes (i,j) so as to

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